

Mine Rescue Rover · 4WD Drivetrain (under the base plate) · rev B

Rev B puts a hall encoder behind every motor. Four identical corner units (mirrored left/right): wheel -> machined hex drive adaptor -> 10 mm steel stub axle -> 3D-printed bearing flange (2 bearings) -> 3D-printed coupler -> 36 mm planetary gear motor **with a rear hall encoder**, held in a 3D-printed U-bracket. Axles run parallel to the 30 cm side on Y = 90 (front) and Y = 310 (rear). The encoders feed the Tier 1 Nano (sheet 8), which now closes the speed loop and sends wheel odometry to the Pi 5 alongside the LIDAR SLAM. Every dimension below comes from one model that is checked by script (sheet 6: **0 failures**). Wheels are **Ø134.5 x 67 1/10** monster-truck wheels (B8), which raise the whole rover by 7.75 mm for ground clearance.

What changes in rev B

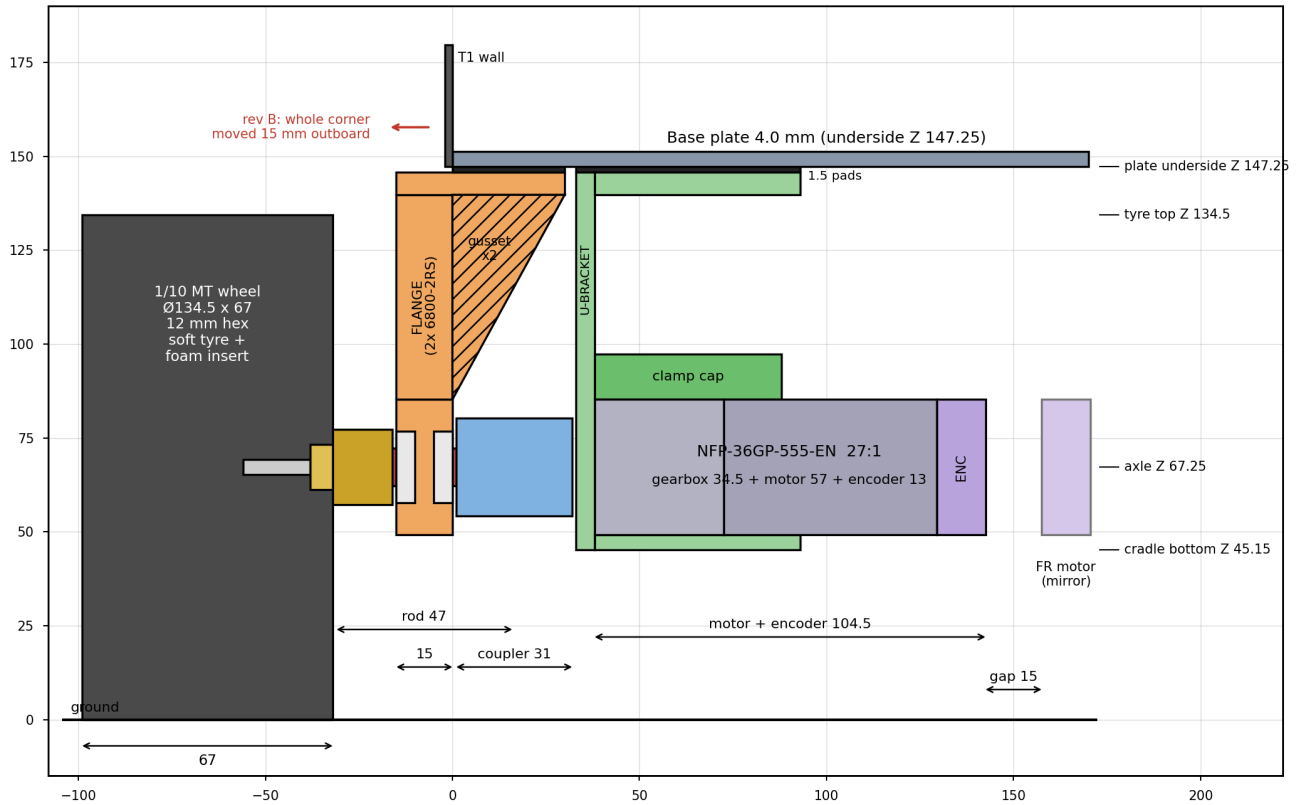
#	Rev A	Rev B	Why
B1	NFP-36GP-555 12 V 27:1, no encoder; body 94.5	NFP-36GP-555-EN 12 V 27:1 : same maker, gearbox, 8 mm D-shaft, face holes, speed (220 rpm no-load) and torque (12.5 kg-cm rated); 17 PPR AB hall encoder on the motor rear. Body 104.5	Gives wheel odometry and a closed speed loop. Only the length changes (+10 mm), so the rest of the corner and the 12 V side stay valid
B2	Motor face X 53, rear X 147.5, 5 mm between opposite motors	Whole corner unit 15 mm outboard : motor face X 38, encoder rear X 142.5, 15 mm between opposite encoders	Two 104.5 motors face to face do not fit in rev A (they would overlap 15 mm). 15 mm also leaves room for the encoder plug
B3	Flange top plate 30 long, boss flush with the plate edge	Flange top plate 45 (X -15 to 30); web and boss 15 mm outside the plate edge; 2 gussets 30 x 54.5 under the plate. Plate bolt holes unchanged (X 8 / 22)	The only part to reprint. Moves the wheel out 15 mm and reuses the rev A flange holes
B4	U-bracket, clamp cap, coupler, axle, adaptor, bearings, shims, pads, bolts	Unchanged parts , all 15 mm further out. Bracket plate holes move from X 63 / 93 to X 48 / 78 (still match-drilled)	Everything already printed or machined is reused
B5	2 grommets Ø12 at (150, 115), (150, 285)	Same, + 2 grommets Ø10 at (150, 65), (150, 335) for the encoder cables	Keeps the 5 V encoder signals away from the PWM motor leads
B6	Track 376, overall width 418	Track 431, overall width 498	Consequence of B2 and B8. Wheelbase/track drops 0.59 -> 0.51: slightly easier skid-steer turns, more roll stability on rubble
B7	Odometry from LIDAR SLAM only	1836 counts per wheel turn (4.55 counts/mm); per-side speed loop, stall cut-out, slip and encoder-fault detection, wheel odometry to the Pi (sheet 8)	Straight driving under uneven load, protection for gearboxes and printed parts, better SLAM
B8	1/10 RC wheel Ø119 x 42, 12 mm hex	1/10 MT wheel Ø134.5 x 67, 12 mm hex : same hex adaptor, axle and corner; the whole rover sits 7.75 mm higher	Ground clearance on rubble: lowest point 37.4 -> 45.15 (42.15 loaded)

Design decisions kept from rev A

#	Old spec	This design	Reason
2	Rod 50 = 15 / 15 / 20	Rod 47 = 15 adaptor / 1 shim / 15 bearings / 1 shim / 15 coupler	Coupler engagement 15 each side; shims keep spinning parts off the printed flange
3	Standoff block + zip tie through the plate	U-bracket cradle + clamp cap hold the motor along 60 mm	Removes the cantilever without holes under the batteries
4	Wheel Ø120	1/10 MT wheel Ø134.5 x 67, 12 mm hex (2.8" monster-truck set)	Real, stocked part with the right hex drive; axle Z 67.25, drop still exactly 80.0
7	Rigid wheel, no damping	Soft-compound tyre with open-cell foam insert , Ø134.5 x 67	The tyre squashes ~3 mm under load and absorbs shock before it reaches the gearbox
8	Flange bolted straight to the plate	1.5 mm neoprene pad (60 Shore A) under the flange AND under the bracket lips ; both printed 1.5 mm shorter	Damps vibration. Both get a pad so the axle and motor stay exactly coaxial
9	Bolt heads on the base plate top	M4x20 countersunk (DIN 7991), heads flush	Nothing pokes up under Battery C, the regulator or the Tier 1 mat
6	Motor L-bracket at X 30 (fixed)	Bracket lips have ±5 mm slots ; plate holes drilled at the slot centres after alignment	Build independent of small motor tolerances (match-drill)

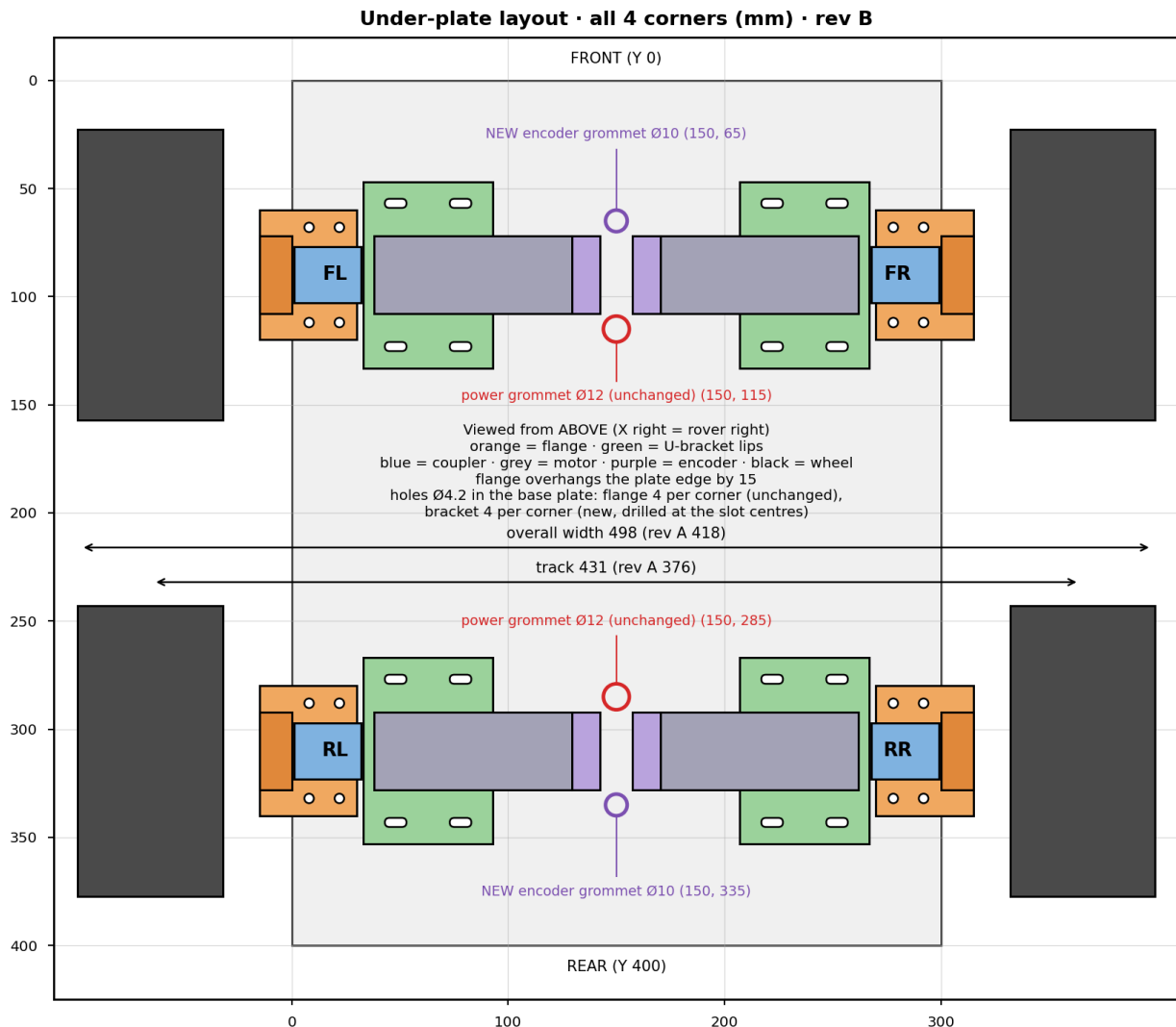
Sheet 1 · Corner geometry

Front-left corner · section on the axle line (Y 90) · X from the left plate edge, Z from the ground · mm · rev B



Part	X from - to (left corner; right = 300 - X)	Z / section
Wheel (tyre)	-99.0 to -32.0 (rev A -59 to -17)	Ø134.5 x 67, centre Z 67.25, top Z 134.5 (12.75 below the plate)
Hex adaptor	stud -56 to -38 · hex -38 to -32 · collar -32 to -16	collar Ø20, hex 12 A/F
Shim (outer)	-16 to -15	10 x 14 x 1 steel shim, rides on the bearing inner race
Flange boss	-15 to 0 (15 outside the plate; inner face flush with the plate edge)	Ø36, 2 x 6800-2RS
Flange top plate	-15 to 30, Y ±30 about the axle; bolts X 8 / 22	6 thick, Z 139.75-145.75; pad X 0-30 only, Z 145.75-147.25
Flange gussets (2)	0 to 30 (under the plate)	5 thick at Y ±10 to ±15 about the axle, Z 85.25-139.75
Shim (inner)	0 to 1	
Stub axle	-31 to 16	Ø10 x 47
Coupler	1 to 32	Ø26
U-bracket front wall	33 to 38	shaft hole Ø14
U-bracket cradle / lips	33 to 93	lips Y ±23.1 to ±43.1 about the axle, Z 139.75-145.75, pad above
Gearbox / can / encoder	38 to 72.5 / 72.5 to 129.5 / 129.5 to 142.5	Ø36, Z 49.25-85.25
Clamp cap	38 to 88	Z 85.35-97.35
Gap to the opposite motor	142.5 to 157.5	15 mm: room for the encoder plugs (sheet 5, step 5)

Sheet 2 · Under-plate layout



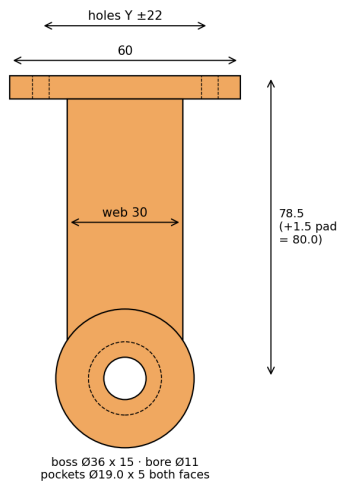
Base plate holes per corner (left front shown; right X -> 300 - X, rear Y -> 400 - Y): **flange** 4 x Ø4.2 with 90° countersink to Ø8.5 on the TOP face, at X 8 and 22, Y 68 and 112 - **unchanged from rev A, reuse them.** **Bracket** 4 x Ø4.2 (same countersink) at the slot centres, nominal **X 48 and 78**, Y 56.9 and 123.1, drilled after the alignment step (rev A nominal X 63 / 93). **Grommets:** Ø12 at (150, 115) and (150, 285) unchanged; **new Ø10 at (150, 65) and (150, 335)** for the encoder cables.

If the rev A bracket holes are already drilled: they sit under the new lips (X 63) or at the lip end (X 93) and 15 mm from the new holes, so the Ø8.5 countersinks do not overlap. Leave them empty, or fill them with an M4 countersunk screw and nut so nothing snags the mat.

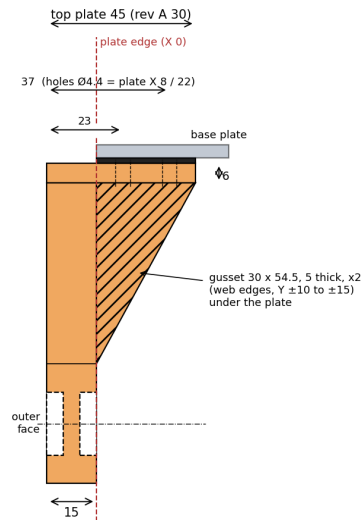
Sheet 3 · Parts

Reprint: **flange only** (4 off). The side view shows what changed: the top plate grows from 30 to 45 so the web and boss hang 15 mm outside the plate edge, and the two gussets are lengthened to 30 so they sit under the plate and carry the overhang into the bolts.

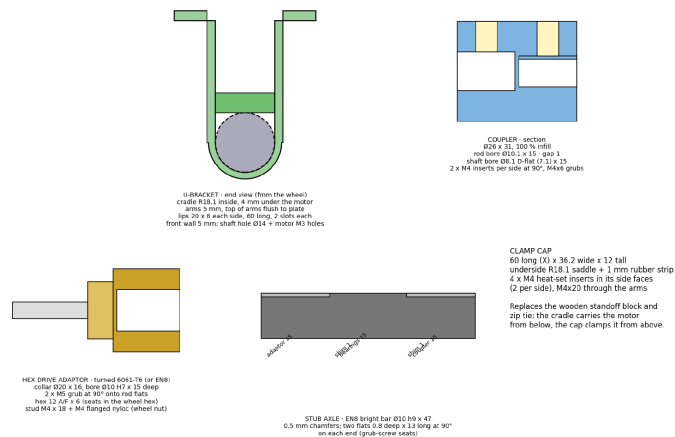
FLANGE · view along the axle (unchanged)



FLANGE rev B · side view (along Y) · X from the boss outer face



Unchanged parts (rev A drawings; reuse rev A prints and machined parts)



3D-printing settings

Part	Qty	Material	Orientation on the bed	Settings
Flange rev B	4	PETG (upgrade: PA6-CF or PETG-CF, hardened nozzle)	Outer boss face down: web, boss and gussets print as in-plane layers, bearing pockets print as true circles; the 45 mm top plate stands vertical	0.4 nozzle, 0.2 layer, 6 walls, 6 top / 6 bottom, 60 % gyroid, 240 °C / bed 80 °C, supports only under the top-plate bolt holes. About 60 g each
U-bracket	4	same	reuse rev A	-
Clamp cap	4	same	reuse rev A	-
Coupler	4	PETG or PA6-CF	reuse rev A	-
Test ring	1	same	flat	Only if the flange material or printer changed: Ø19.0 pocket ring, bearing must press in by hand-vise

Printed holes are drawn at 4.4 mm (not 4.2) because printed holes shrink about 0.2 mm; drill to 4.2-4.4 if tight. Metal plate holes stay 4.2 mm.

Sheet 4 · Procurement (for 4 corners)

Item	Exact part	Qty	Note
Motor (changed)	NFP-36GP-555-EN 12 V, 27:1 (gearbox 34.5 mm), 17 PPR AB hall encoder, XH2.54 6-pin cable, Ø8 D-shaft. Same size alternative: CHIHAI CHP-36GP-555-ABHLL 27:1	4	NFP lists it at about US\$43. Ask for the outline drawing: body (gearbox face to encoder cover) must be 104.5 (max 106). Shaft extension 17-21 mm; cut to 21 if longer. Match-drill the M3 face holes
Wheels (changed)	1/10 MT 2.8" wheel + tyre, 12 mm hex, Ø134.5 x 67 , pre-glued, e.g. Jetko Rockform JK2803CB (ships with 12 / 14 / 17 mm hexes)	2 pairs	Fit the 12 mm hex. Prefer soft compound and an open-cell foam insert. Check the M4 wheel stud still takes the full nyloc through the hub
Grommets (new)	Rubber, 10 mm hole	2	Encoder cables, (150, 65) and (150, 335)
Encoder wiring (new)	4-core 24 AWG flexible cable (or 4 twisted wires) + JST XH2.54 6-pin pigtail if the motor cable is short; Dupont or screw terminals at the Nano	4 x ~50 cm	Heat-shrink every joint; label FL / RL / FR / RR at both ends
Filament	PETG (or PA6-CF / PETG-CF)	~0.3 kg	4 new flanges
Bearings	6800-2RS, 10 x 19 x 5	8	reuse rev A
Stub axles, hex adaptors	as rev A	4 / 4	reuse rev A
Pads	Neoprene 1.5 mm, 60 Shore A	8 pads	Lip pads need new holes at the new bracket-hole positions: cut fresh 60 x 20 pads (x 8) if the old ones are punched in the wrong place
Shims, bolts, cap bolts, inserts, grubs, wheel nuts, motor screws	as rev A	as rev A	reuse rev A
Grommets	Rubber, 12 mm hole	2	unchanged (motor leads)

Rev A motors (no encoder) cannot take a rear encoder: their can has no rear shaft to carry the magnet ring. Keep them as spares for other mechanisms. The Ø119 x 42 wheels are replaced by the Ø134.5 x 67 set; keep them as spares too.

Sheet 5 · Assembly (base plate upside down on the bench)

1 · Prepare

Print the 4 rev B flanges. Drill the two new Ø10 grommet holes (sheet 2); the flange holes and Ø12 grommets are unchanged. Leave the bracket holes for step 6.

2 · Flanges

Press two 6800-2RS into each flange (one per face, flush). Lay the 1.5 mm pad (60 x 30) where the top plate meets the base plate (X 0-30), then bolt: countersunk head flush in the plate top, plate, pad, flange, rubber washer, nyloc. Tighten until the pad just starts to bulge. The boss now hangs outside the plate: its **inner** face is flush with the plate edge, the outer face is 15 mm out.

3 · Axles

As rev A: outer shim, axle through the bearings from outside, 15 mm outboard beyond the shim and 16 mm inboard. Fit the inner shim.

4 · Couplers

As rev A: coupler rod bore fully onto the inner 15 mm, against the inner shim; grubs onto the flats (Loctite 243).

5 · Motors and encoder socket

Find the encoder socket on the motor rear. **Radial socket:** turn the motor so it points toward the nearest plate end (front motors to the front, rear motors to the rear), i.e. toward the new encoder grommet. **Axial socket** (on the back face): left motors socket at 12 o'clock, right motors at 6 o'clock, so the two plugs in the 15 mm gap never face each other. The M3 face holes are match-drilled, so any rotation works. Then screw the face to the U-bracket wall (M3), lay the motor in the cradle with the rubber strip, fit the clamp cap and its 4 bolts.

6 · Align and drill (match-drill)

As rev A: slide the shaft into the coupler until the lips sit flat and the wall is 1 mm from the coupler end; mark the 4 slot centres (nominal X 48 / 78), drill Ø4.2, countersink the top, refit with the lip pads and bolt. Tighten the coupler shaft grubs. Turn the wheel end by hand: no binding.

7 · Wheels

Fit the 12 mm hex in each new wheel first (the set also ships 14 and 17 mm hexes). Then as rev A: hex adaptor on the outer 15 mm against the outer shim, M5 grubs (Loctite 243), wheel on the hex, M4 flanged nyloc.

8 · Wiring

Motor leads up through the Ø12 grommets: FL + RL to the LEFT driver M+ / M-, FR + RR to the RIGHT driver (Part 1). Encoder cables up through the Ø10 grommets to the Tier 1 Nano (sheet 8). If the motor's 6-pin plug also carries M+ / M-, split it at the motor: those two go to the driver on 18 AWG, only the four encoder wires go to the Nano. Zip-tie the cables to the lips, clear of the tyres. Do not swap M+ / M- to fix direction any more: the bring-up check stores the signs in the Nano.

9 · Check

On the floor the tyres squash about 3 mm; lowest point (cradle) about 42.2 mm; 15 mm between opposite encoder covers. Then run the bring-up check (sheet 8, step 3) with the wheels off the ground.

Sheet 6 · Verification

	Check (script: drivetrain_model_revB.py)
OK	Flange/bracket drop plate-underside to axle = 80.0 mm (spec 80.0)
OK	Axle height = tyre radius = 67.25 mm
OK	Tyre top 134.5, plate underside 147.25 -> clearance 12.75 mm
OK	Rod 47 mm = adaptor 15 + shim 1 + bearings 15 + shim 1 + coupler 15 (unchanged part)
OK	Coupler end 32.0, bracket wall starts 33.0 (gap 1.0)
OK	Motor shaft engagement in coupler 15.0 mm (min 12)
OK	Coupler clears flange boss (shim 1.0 mm between)
OK	Motor body = gearbox 34.5 + can 57 + encoder 13 = 104.5 mm
OK	Left motor rear 142.5, right motor rear 157.5 -> gap 15.0 mm (≥ 12 for an axial encoder socket)
OK	Flange top plate ends X 30, bracket lips start X 33 (gap 3.0)
OK	Flange top plate X -15 to 30: 15 mm overhang past the plate edge, bolts stay at X 8 / 22 (rev A holes reused)
OK	Gussets (under the plate) clear the coupler top Z 80.25 by ≥ 6.8 mm
OK	Gusset ends X 30, U-bracket wall starts X 33 (gap 3.0)
OK	With 3 mm soft-tyre squash under load: lowest point 42.15 mm
OK	Lowest point (bracket cradle) 45.15 mm above ground; motor + encoder 49.25; flange boss 49.25
OK	Tyre inner face at X -32.0 -> 17.0 mm to the flange boss, 32 mm outside the plate edge
OK	Track (tyre centre to centre) 431 mm, overall width 498 mm (rev A 418)
OK	Tyres along Y: front 22.75-157.25, rear 242.75-377.25 -> inside the plate length, 85.5 mm apart
OK	M4x20 countersunk flange (CSK flush) + 1.5 pad: plate 4 + pad 1.5 + part 6 + rubber washer 1 + nyloc 5 -> 2.5 mm past nut
OK	M4x20 countersunk bracket lip (CSK flush) + 1.5 pad: plate 4 + pad 1.5 + part 6 + rubber washer 1 + nyloc 5 -> 2.5 mm past nut
OK	Pads: flange and bracket lips both on 1.5 mm pads, both printed 1.5 mm shorter -> axle and motor coaxial at 80.0 below the plate
OK	M4x20 arm into cap insert: 14.2 mm of thread in the cap (clamp cap unchanged)
OK	Under-plate footprints: flange top, bracket lips, 4 grommets - no overlaps
OK	Encoder grommet (150, 65): its edge is 2 mm off the motor side in plan and 62 mm above the motor top
OK	Motor cantilever past the cradle: 49.5 mm (rev A 39.5); clamp cap and U-bracket unchanged
OK	Encoder: 17 PPR $\times$ 27:1 $\times$ 4 = 1836 counts per wheel turn = 4.55 counts/mm (loaded tyre Ø128.5)
OK	Worst-case edge rate, 4 motors at no-load 220 rpm: 26.9 k interrupts/s ($\sim 5 \mu\text{s}$ each -> 13 % of the Nano)
	TOTAL FAILURES: 0

Confirm on the real parts

#	Check
1	Motor shaft extension (face to tip) is 17-21 mm; cut to 21 if longer. With 21 the coupler takes 15 mm of shaft.
2	Motor body length (gearbox face to the back of the encoder cover, socket not included) is 104.5, at most 106. Above 106 the gap between opposite motors drops below 12 mm. Measure one motor before printing the flanges.
3	Encoder socket: if it is axial, plug + wire bend must stick out 11 mm or less from the cover (sheet 5, step 5 staggers the plugs).
4	Encoder pinout from the motor label / datasheet: which wires are M+, M-, GND, VCC, A, B. VCC takes 3.3-5 V only, never 12 V.
5	Chassis doc: nothing hangs under the plate where the lips now sit (X 33-93 and 207-267, Y 46.9-133.1 and 266.9-353.1), in particular no lower-T1-bracket nut or bolt tip, and nothing hangs below the plate edge at the flange overhang (X -15 to 0). This model does not contain the chassis bolts that rev A's check used.
6	Tier 1 layout above: the new Ø10 holes at (150, 65) and (150, 335) and the bracket holes at X 48 / 78 are clear of everything on the plate top (standoffs, pillars, battery bays).
7	Wheel: 12 mm hex seat; the tyre inner face ends up 32 mm outside the plate edge, and the M4 stud still takes the full nyloc through the new wheel's hub.
8	Bearing pocket press fit on the test ring (only if the flange material or printer changed).
9	After fitting the pads, the axle and motor turn freely by hand; if the coupler binds, the two pads are squeezed unequally.

Sheet 7 · Insulation mat (chassis, all decks)

A mat on every enclosed deck floor keeps loose wires, solder joints and bolt tips off the aluminium. It goes inside each deck only, not on the exposed ledges of the lower roofs and not on the top roof.

Item	Specification
Material	Silicone rubber sheet, 1.5 mm, 50 Shore A, flame-retardant grade (UL94 V-0), rated to 200 °C. Silicone survives the heat near the BTS7960 heatsinks and does not melt onto the plate
Tier 1	30.0 x 40.0 cm on the base plate
Tier 2	22.0 x 35.0 cm on the mid plate, inside the Tier 2 box
Tier 3	18.1 x 31.1 cm on the Tier 2 roof, inside the Tier 3 box
Cut-outs (+1 mm all round)	Pillars, every lip / lower bracket, latch posts and the wire pass-through holes: brackets must clamp metal to metal, never on the mat. Rev B: add Ø12 cut-outs for the two new encoder grommets at (150, 65) and (150, 335)
Cut-outs under breadboards	All 400-pt and 170-pt breadboards stick straight onto the aluminium (their adhesive will not hold on silicone). This keeps the MQ and MLX90640 heights, so the gas ports (4.40 cm) and thermal arches (3.5 cm) are unchanged
Holes	Punch Ø5 for every bolt and standoff that passes through the floor (drivers, bus bars, Pi 5, hub, battery bays)
Fixing	Held down by the parts bolted through it; add a few dots of neutral-cure RTV silicone at the corners
Height effect	Parts sitting on the mat rise 1.5 mm. The tallest (Tier 1 drivers, 4.5 cm) become 4.65 cm, still 2.8 cm below the nearest bolt tip hanging from the mid plate (7.42 cm). All drivetrain bolt heads are countersunk flush, so nothing pushes up under the mat

Quantity: one 50 x 50 cm sheet covers Tier 1; one more sheet covers Tier 2 and Tier 3.

Sheet 8 · Encoders and the Tier 1 Nano

Each motor carries a 2-channel hall encoder: 17 pulses per motor turn on A and B, 90° apart. Through the 27:1 gearbox with x4 decoding that is **1836 counts per wheel turn, 4.55 counts per mm** (loaded tyre Ø128.5). The Tier 1 Nano counts all four, runs a 50 Hz speed loop per side (the two motors of a side share one BTS7960) and streams the counts to the Pi 5 at 25 Hz. Worst case (all four at 220 rpm no-load) is 26.9 k edges/s, well inside the Nano's interrupt budget.

Encoder cable (per motor)

Encoder wire	Goes to	Note
VCC	Tier 1 5 V rail (Battery B)	3.3-5 V only. The encoder has its own pull-ups, so A / B swing 0-5 V
GND	Tier 1 GND trunk	No extra link to the 12 V side (one common GND link rule, Part 2 sheet C)
A, B	Nano pins in the table below	Keep these 4 wires together, away from the motor leads
M+, M- (if on the plug)	BTS7960 M+ / M- of that side, 18 AWG	Split at the motor, through the Ø12 grommet (sheet 5, step 8)

Tier 1 Nano pin map

Nano pin	Connects to	Why
D9 / D10	LEFT BTS7960 RPWM / LPWM	Timer1 PWM, 3.92 kHz (driver limit 25 kHz)
D8	LEFT BTS7960 R_EN + L_EN (tied)	HIGH = enabled; LOW while booting
D3 / D11	RIGHT BTS7960 RPWM / LPWM	Timer2 PWM, same 3.92 kHz
D12	RIGHT BTS7960 R_EN + L_EN (tied)	
D4 / D5	FL encoder A / B	Port D pin-change interrupt (fixed pins)
D6 / D7	RL encoder A / B	Port D pin-change interrupt (fixed pins)
A0 / A1	FR encoder A / B	Port C pin-change interrupt (fixed pins)
A2 / A3	RR encoder A / B	Port C pin-change interrupt (fixed pins)
D13	on-board LED	slow blink running · solid waiting for the Pi · fast blink fault
5V / GND	Tier 1 5 V rail: both drivers' VCC / GND and the 4 encoders' VCC / GND	Part 2 sheet C
USB (D0 / D1)	Pi 5 through the hub	115200 baud; free pins: D2, A4, A5, A6, A7

Driver pins match the Part 1 wiring sheet. Driver pins can be changed in the constants at the top of the sketch (PWM must stay on D3 / D9 / D10 / D11). The encoder pins are fixed: the interrupt code reads whole ports. BTS7960 R_IS / L_IS stay unconnected; stalls are detected from the encoders.

Link to the Pi 5 (full description in the sketch header)

Direction	Message	Meaning
Pi -> Nano	V, left, right	wheel speed per side, mm/s (±1000), closed loop
	P, left, right	open-loop PWM (±255), bring-up only
	S · R · I · D	brake · zero counters, clear faults · identify · dump config
	C, fl, rl, fr, rr, ml, mr · K, kp, ki, kff	signs (±1) · gains x1000, both saved in EEPROM
Nano -> Pi	E, ms, fl, rl, fr, rr, vl, vr, pl, pr, flags, errs* HH	25 Hz: cumulative counts, side speeds, PWM, flags (stall, slip, encoder fault, watchdog), missed edges
	I, TIER1_DRIVE, 4B, 1.0 · A, x · N, reason	identity (at boot) · ack · reject

Built-in protection

Situation	What the Nano does
No command for 300 ms	Both sides ramp to 0 and brake (watchdog flag)

Situation	What the Nano does
Stall: PWM $\geq$ 140 with the side below 15 mm/s for 0.7 s	That side is braked and latched until the Pi commands 0 for it (after 1 s) or sends R. Protects gearboxes, printed couplers and the 20 A fuse
Any speed change	Setpoint ramp 1200 mm/s ²
One encoder dead (its partner turns, it does not)	Flag for that wheel; the side keeps running on the other wheel's encoder
Front and rear of one side differ ($>$ 100 mm/s or 30 %)	Slip flag; the ROS 2 node widens the odometry covariance

Bring-up and calibration

Step	Do this
1	Flash tier1_drive_nano/tier1_drive_nano.ino (Arduino IDE, board Arduino Nano, ATmega328P; pick 'Old Bootloader' if the upload times out). The LED stays on: waiting for the Pi.
2	On the Pi 5: <code>python3 -m pip install pyserial</code> , then <code>ls /dev/serial/by-id/</code> to find the Tier 1 Nano (the Tier 2 Nano and the ESP32 are on USB too; always use the by-id path).
3	Rover on blocks, wheels off the ground, 12 V on: <code>python3 rover_base.py --port <path> --check</code> . It spins each side slowly, asks whether the wheels turned forward, and stores the motor and encoder signs in the Nano. A reversed side is fixed in software.
4	On the floor: <code>--calib-distance 1000</code> (tape measure), then <code>--calib-turn</code> (one turn in place). Results go to <code>~/.config/rover_base.json</code> . Skid steer usually calibrates to an effective track larger than 431.
5	<code>--monitor</code> shows counts, speeds and flags. For the SLAM stack: <code>python3 rover_base_ros2.py --ros-args -p port:=<path> (/cmd_vel in; /odom and the odom -> base_link TF out)</code> .

Simulation test (simulated Nano with motor, quadrature and driver models, driven by the Pi code): sign detection on deliberately mis-wired hardware, 300 mm/s tracking, straight and turning odometry, stall latch and release, slip, dead-encoder fallback, watchdog and config storage all passed. Final gains are tuned on the real rover with `--gains`.